Topic 1: Data Communication & Networking Fundamentals

Course outline weeks 1-2: introduction concepts, goals & applications, network structure, topology design. Appears in EVERY past paper.

How often this topic is examined

Question asked	Paper(s)	Marks
Components of a data communication system (with diagram)	2018 Q1a, SE2024 Q2f	5
Transmission modes: simplex / half duplex / full duplex	2023 Q4e, SE2024 Q1a	1-6
Categories of networks (LAN, MAN, WAN...)	2018 Q1c, ISA2024 Q3b	6-9
Network topologies (describe 3 or 4, ring vs bus)	2025 Q2c, ISA2024 Q2c, SE2024 Q2e	3-12
Logical vs physical topology	SE2024 Q3c	2
Client-server vs peer-to-peer	2025 Q1b, SE2024 Q4b/Q5d	4-8
What is a protocol (with examples)	2025 Q2a	5
Point-to-point communication / configurations of networks (relationships among nodes)	SE2024 Q1a, Q3d	1-4
Internet vs WWW; internet services	SE2024 Q4a, Q5c	4
Hardware, software & protocols of a basic network	SE2024 Q4d	4
Network design scenario (5 floors x 20 PCs)	SE2024 Q3e	5
Competencies: network admin, engineer, hacker	SE2024 Q4e	4

EXAM TIP: Topologies + components + client/server are 'free marks' questions. Always DRAW a diagram even when not asked - the 2024 ISA paper says 'illustrate your answer with large clearly labelled diagrams'.

1. Components of a data communication system

EXAM QUESTION 2018 Q1a / SE2024 Q2f **[5]**
With the aid of a fully labelled diagram, explain all the components involved in a data communication system.

Made by King Tine, the AI Engineer

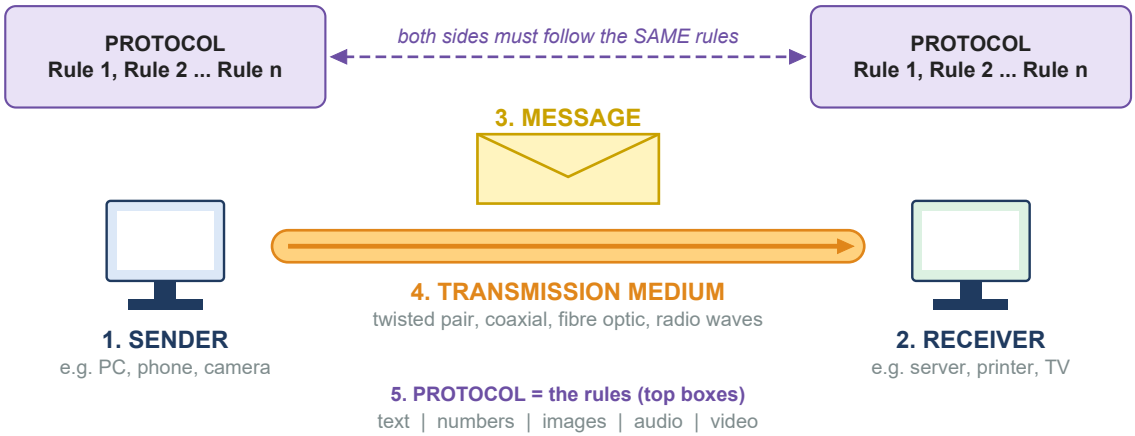


Figure: the 5 components of a data communication system

1. **Message** - the information (data) to be communicated: text, numbers, pictures, audio or video.
2. **Sender** - the device that sends the message, e.g. computer, phone, video camera.
3. **Receiver** - the device that receives the message, e.g. computer, printer, TV.
4. **Transmission medium** - the physical path the message travels: twisted-pair, coaxial, fibre-optic cable or radio waves.
5. **Protocol** - a set of rules that govern the communication. Without a protocol two devices may be connected but not communicating (like a Shona speaker and a French speaker).

REMEMBER: 5 components = **M-S-R-T-P**: Message, Sender, Receiver, Transmission medium, Protocol. Characteristics of a good system: **Delivery, Accuracy, Timeliness, Jitter**.

2. Transmission modes (data flow)

EXAM QUESTION 2023 Q4e **[6]**
The transmission mode defines the direction of signal flow between two connected devices. State the three transmission modes, giving examples for each.

Mode	Meaning	Example
Simplex	One direction only. One device transmits, the other only receives.	Keyboard -> computer, TV/radio broadcast, monitor
Half duplex	Both directions but **not at the same time**. Whole channel capacity used by whichever device is sending.	Walkie-talkie / two-way radio, CB radio
Full duplex	Both directions **simultaneously**. Channel capacity is shared (or two separate paths).	Telephone call, mobile phone, video call

Half duplex transmission (SE2024 Q1a.i, 1 mark): communication in both directions but only one direction at a time, e.g. walkie-talkie.

Point-to-point communication (SE2024 Q1a.ii): a dedicated link between exactly two devices; the whole capacity of the link is reserved for those two, e.g. a TV remote and TV, or a leased line between two routers. Opposite = **multipoint (multidrop)**, where more than two devices share one link.

Synchronous transmission (SE2024 Q1a.iii): data sent as a continuous stream of bits in blocks/frames with sender and receiver clocks synchronised (timing signal), no start/stop bits per byte. Faster/efficient. Opposite is **asynchronous** - each byte framed by start and stop bits, gaps between bytes allowed (e.g.

keyboard).

3. Categories / types of networks

> **EXAM QUESTION** 2018 Q1c [6], ISA2024 Q3b [9] **[6-9]**
> There are various categories of networks. Explain any three. / Describe the differences between LAN, WAN and MAN.

Feature	LAN	MAN	WAN
Full name	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Single room, building or campus (up to ~ a few km)	A city / town (~5-50 km)	Country, continent, the whole world
Ownership	Privately owned by one organisation	Usually a company or municipality / ISP	Public carriers, telcos (e.g. TelOne, Liquid), leased
Speed	Very high (100 Mbps - 10 Gbps)	High	Lower than LAN, depends on link
Error rate / delay	Low errors, low delay	Moderate	Higher errors and propagation delay
Cost / setup	Cheap, easy to set up	Expensive	Very expensive
Technology	Ethernet, Wi-Fi	Fibre rings, WiMAX, cable TV	MPLS, leased lines, satellite, the Internet
Example	HIT computer lab	Network linking Harare city council offices	The Internet, bank branches across Zimbabwe

Others you can mention: **PAN** (Personal Area Network, ~10 m, Bluetooth), **CAN** (Campus Area Network), **SAN** (Storage Area Network), **WLAN** (wireless LAN).

4. Network topologies

> **EXAM QUESTION** 2025 Q2c [12] (also ISA2024 Q2c - three topologies [9]) **[12]**
> Describe any four (4) network topologies that can be used to connect devices.

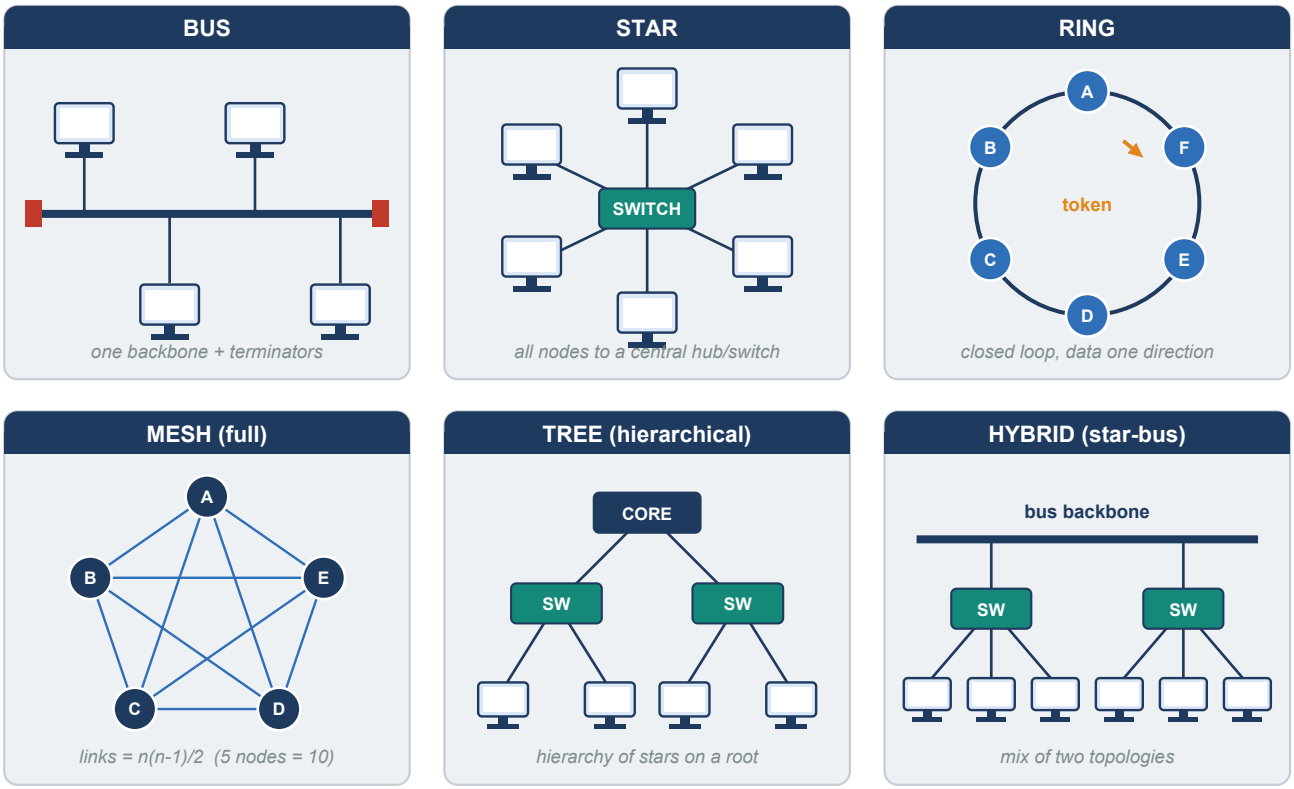


Figure: the six network topologies

Topology	How it works	Advantages	Disadvantages
Bus	All devices attach to one backbone cable with terminators at both ends; a signal is broadcast to every node.	Cheap, little cable, easy to extend	Backbone fault brings network down; collisions; hard to troubleshoot; performance drops as nodes increase
Star	Each device has its own cable to a central hub/switch.	Easy install & fault isolation; one cable failure affects only one node; easy to add nodes	Central device is a single point of failure; more cable than bus
Ring	Each device connects to two neighbours forming a closed loop; data passes in one direction, usually using a token.	No collisions (token); equal access; good under heavy load	One break/one faulty node can stop the whole ring; adding/removing nodes disrupts network; slower (each node repeats data)
Mesh	Every device has a dedicated point-to-point link to every other device. Needs n(n-1)/2 links.	Very reliable/fault tolerant; dedicated links = no traffic problems; secure/private	Very expensive; lots of cable & I/O ports; hard to install
Tree	Hierarchy of star networks connected to a backbone.	Scalable, easy to manage in sections	If the root/backbone fails, segments are cut off
Hybrid	Combination of two or more topologies (e.g. star-bus, star-ring).	Flexible, uses strengths of each	Complex design, expensive

EXAM QUESTION SE2024 Q2e **[3]**
Explain the limitations of a ring network topology compared to a bus network topology.

- A single break in the cable or failure of one node can break the whole ring (unless a dual ring is used); in a bus, a failed node does not affect others.

- Adding or removing a device requires breaking the ring, which disrupts the network; in a bus you simply tap into the backbone.
- Every node must receive and regenerate the data, so delay grows with the number of nodes; it is also more expensive/complex (token management) than a bus.

Logical vs physical topology (SE2024 Q3c.i, 2 marks)

Physical topology = the actual physical layout of cables and devices (how it looks). **Logical topology** = how data actually flows between devices (how it works). Example: Ethernet using a switch is a physical **star** but logically behaves like a **bus**; Token Ring wired to a MAU is a physical star but a logical **ring**.

5. Network configurations by relationship among nodes

> **EXAM QUESTION** SE2024 Q3d **[4]**
> Describe in detail four possible configurations of computer networks as categorized by relationships among nodes.

1. **Peer-to-peer** - every computer is equal; each can act as client and server and share its own resources. No central server.
2. **Client-server** - dedicated servers provide services (files, mail, web, database) to clients that request them.
3. **Point-to-point** - a dedicated link joins exactly two nodes.
4. **Multipoint / broadcast** - many nodes share one link/channel; a message sent by one is received by all (e.g. bus, Wi-Fi).

(If the lecturer means line configuration only, point-to-point and multipoint are the two main ones; give the client/server and P2P as the 'relationship' models.)

6. Client-server vs peer-to-peer

> **EXAM QUESTION** 2025 Q1b [8]; SE2024 Q4b - four benefits of client-server [4] **[8]**
> With the aid of proper examples, differentiate between the Client-Server model and the Peer-to-Peer model for network applications.

Point	Client-Server	Peer-to-Peer (P2P)
Structure	Central, powerful server(s) serve many clients	All nodes are equal peers; each is both client and server
Control & security	Centralised administration, user accounts, access control	Each user manages own machine; weak security
Data storage / backup	Data stored centrally, easy backup	Data scattered across peers, backup difficult
Scalability	Scales well (add servers)	Performance falls as peers grow (for small nets) - but file-sharing P2P scales because each peer adds capacity
Cost	Expensive: server hardware, NOS, admin staff	Cheap, no dedicated server
Failure	Server is a single point of failure	No single point of failure

Point	Client-Server	Peer-to-Peer (P2P)
Examples	Web (browser <-> web server), email (Outlook <-> mail server), HIT e-learning portal, online banking	BitTorrent, Skype (original), Windows workgroup file sharing, blockchain/Bitcoin

Four benefits of client-server over P2P: (1) centralised security & user authentication, (2) centralised backup and data integrity, (3) easier management/updates of resources, (4) better scalability and performance with dedicated servers.

> **EXAM QUESTION** SE2024 Q5d **[7]**
> With reference to the client-server paradigm, explain how Harare Institute of Technology students are able to access online learning resources remotely from home.

1. The student's device (laptop/phone with a browser = **client**) connects to the Internet through their ISP (Wi-Fi/mobile data).
2. The student types the URL of the HIT e-learning portal (e.g. Moodle). The client asks a **DNS** server to resolve the domain name to the server's IP address.
3. The client opens a **TCP** connection (3-way handshake) to the web server on port 443 (HTTPS); TLS encrypts the session.
4. The client sends an **HTTP request** (GET login page). The student logs in; the server **authenticates** the credentials against a database/directory.
5. The **server** processes requests, fetches notes, videos, assignments from its database/file storage and sends **HTTP responses** back.
6. The browser renders the resources; uploads (assignments) travel the opposite way (HTTP POST).
7. The server handles many clients at once; resources are stored centrally so every student sees the same updated material. Firewalls / VPN protect the campus network.

7. Protocols

> **EXAM QUESTION** 2025 Q2a **[5]**
> Protocol is usually associated with the software components of the network, explain with the aid of examples what a protocol is.

A **protocol** is a set of rules that govern data communication - it defines **what** is communicated, **how** it is communicated and **when**. Key elements:

- **Syntax** - structure/format of the data (e.g. first 8 bits = source address).
- **Semantics** - meaning of each section of bits and what action to take.
- **Timing** - when data should be sent and how fast.

Examples: **HTTP/HTTPS** (web pages), **SMTP/POP3/IMAP** (email), **FTP** (file transfer), **TCP/UDP** (transport), **IP** (addressing & routing), **DNS** (name resolution), **DHCP** (automatic IP configuration), **Ethernet** (LAN framing).

Standards: de facto (by convention) vs de jure (by law/official body). Bodies: ISO, ITU-T, ANSI, IEEE, EIA, IETF.

8. Internet vs World Wide Web; Internet services

EXAM QUESTION SE2024 Q4a **[4]**
Distinguish between the internet and the World Wide Web.

Internet	World Wide Web (WWW)
Global network of interconnected networks (hardware + infrastructure)	A service/application that runs ON the Internet
Uses TCP/IP	Uses HTTP/HTTPS, HTML, URLs
Carries email, FTP, VoIP, gaming, WWW...	Collection of linked web pages / hypertext documents
Started 1969 (ARPANET)	Invented by Tim Berners-Lee, 1989-1991

EXAM QUESTION SE2024 Q5c **[4]**
The internet offers a variety of services to users. List and explain any four.

- **World Wide Web** - browsing web pages and web apps (HTTP).
- **Email** - sending/receiving electronic mail (SMTP, POP3, IMAP).
- **File transfer** - uploading/downloading files (FTP, SFTP), cloud storage.
- **Communication** - VoIP calls, video conferencing (Zoom), instant messaging (WhatsApp).
- Others: remote login (Telnet/SSH), e-commerce & e-banking, online learning, social media, streaming.

9. Basic network requirements: hardware, software, protocols

EXAM QUESTION SE2024 Q4d **[4]**
A basic computer network must consist of at least hardware, software and protocols. Give examples of each and explain their implementation.

Requirement	Examples	Implementation / role
Hardware	NICs, cables (UTP, fibre), switches, routers, access points, servers, PCs	Physically connect devices and move signals; switch connects LAN devices, router connects networks
Software	Network OS (Windows Server, Linux), drivers, browsers, email clients, firewall software	Manage resources, users and applications that use the network
Protocols	TCP/IP, Ethernet, HTTP, DHCP, DNS	Rules so devices can understand each other: addressing, delivery, error control

10. Network design scenario

EXAM QUESTION SE2024 Q3e **[5]**
Identify appropriate network topology and network connecting devices. Draw a network design for an organisation having its office in a building of 5 floors. Each floor needs 20 machines.

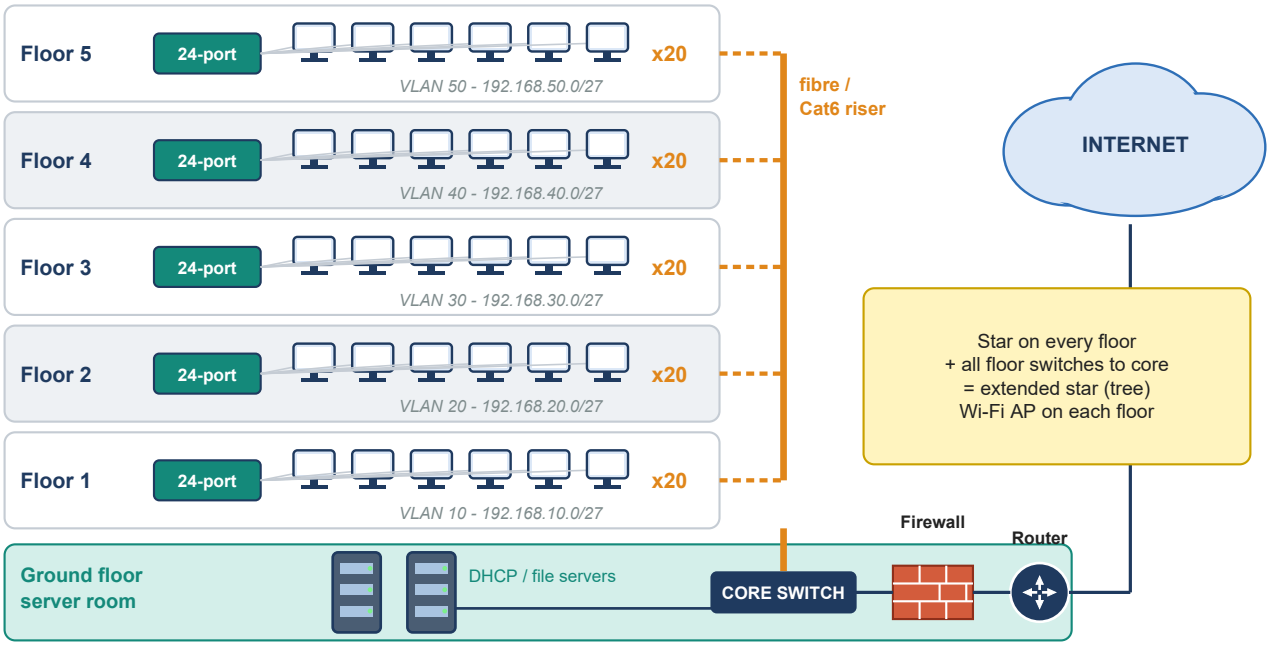


Figure: network design for a 5-floor building, 20 PCs per floor

- **Topology:** extended / hierarchical **star (tree)** - a star on each floor, all floor switches connected to a core switch.
- **Devices:** one 24-port switch per floor (20 PCs + AP + uplink), core switch, router (connects to ISP/Internet), firewall, wireless access points, Cat6 UTP to desks, fibre/Cat6 vertical backbone.
- Use one **VLAN/subnet per floor** (e.g. 192.168.10.0/27 ... 192.168.50.0/27 - each /27 gives 30 hosts for 20 PCs + growth). DHCP server for addressing.
- Why: easy to manage, one failed floor switch only affects that floor, easy to expand.

11. Competencies of professionals (SE2024 Q4e)

- **Network administrator** [1] - day-to-day running: configuring and monitoring the network, user accounts, backups, troubleshooting, patching, keeping it available.
- **Network engineer** [2] - designs and plans networks (topology, addressing, routing), selects and installs equipment (routers, switches, firewalls), optimises performance and security; certifications such as CCNA/CCNP.
- **Hacker** [1] - person with deep technical skills who finds and exploits vulnerabilities in systems; ethical (white-hat) hackers do this legally for penetration testing, black-hat hackers do it maliciously.

12. Goals & applications of networks (outline week 1)

- **Resource sharing** - printers, files, Internet connection.
- **Communication** - email, messaging, video conferencing.
- **Reliability** - multiple copies of data/alternative paths.
- **Cost reduction** - share expensive hardware/software.
- **Centralised management, scalability, remote access.**
- Applications: e-banking, e-learning, e-commerce, cloud services, IoT, telemedicine.

REMEMBER: In a 25-mark question, examiners award ~1 mark per correct point + 1 for an example. 'Explain' = definition + how it works + example. 'Differentiate' = use a TABLE.